AE 358 RENEWABLE ENERGY SYSTEMS AND MANAGEMENT

LECTURE 8

WIND ENERGY

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Learning Objectives

- Explain how wind is formed and the factors that are most important for wind energy production.
- Describe the principal types of wind turbines.
- Discuss wind resource assessment and impacts of wind power projects

Overview of wind energy

- Wind is caused by differences in pressure as a result of temperature differences.
- When a pressure gradient exists, air is accelerated from the point of higher pressure to that of lower pressure.
- Local winds are caused either by topographic effects or variations in surface composition in the immediate area.
- The type of vegetation may have a significant influence through its effects on the absorption or reflection of solar radiation, affecting temperatures and on humidity which in-turn affects pressure which is the main driving force of wind.

Power in the wind

- Wind energy is basically the kinetic energy of the air in motion.
- A wind turbine converts the kinetic energy of the wind into a torque which acts on the turbine blades.
- The amount of energy which the wind transfers to the turbine rotor depends on the density of air, the rotor area and wind speed

$$P = \frac{1}{2}\dot{m}V^2 = \frac{1}{2}(\rho A V)V^2 = \frac{1}{2}\rho A V^3$$

P is power (W), ρ is density (kg/m^3), A is area (m^2) and V is velocity (m/s)

$$\frac{P}{A} = \frac{1}{2}\rho V^3 \; is\ referred\ to\ as\ the\ Wind\ Power\ Density$$

Wind power parameters

- **Density**
  - The kinetic energy of a moving body is proportional to its mass. The kinetic energy in the wind, thus, depends on the density of the air.
- **Rotor Area**
  - The rotor area determines how much energy a wind turbine is able to harvest from the wind. A typical 1 MW wind turbine has a rotor diameter of 54 m (a rotor area of some 2300 m^2)
- **Wind Speed**
  - The energy content of the wind varies with the cube of the average wind speed.

Wind turbines

- Wind turbines are the devices used to extract power from the wind.
- If the mechanical energy is used directly by machinery, such as a pump, it is called a windmill.
- If the mechanical energy is converted to electricity, the machine is called wind turbine, wind generator, wind power unit, wind energy converter or aerogenerator.

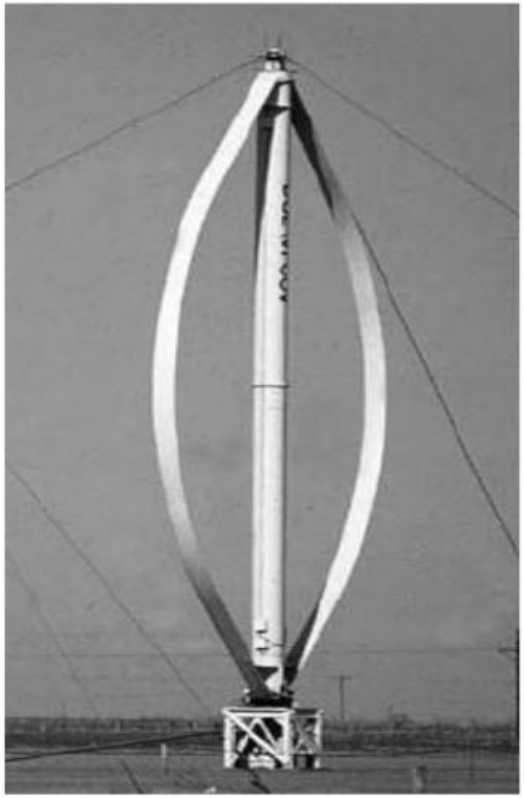
Wind mill



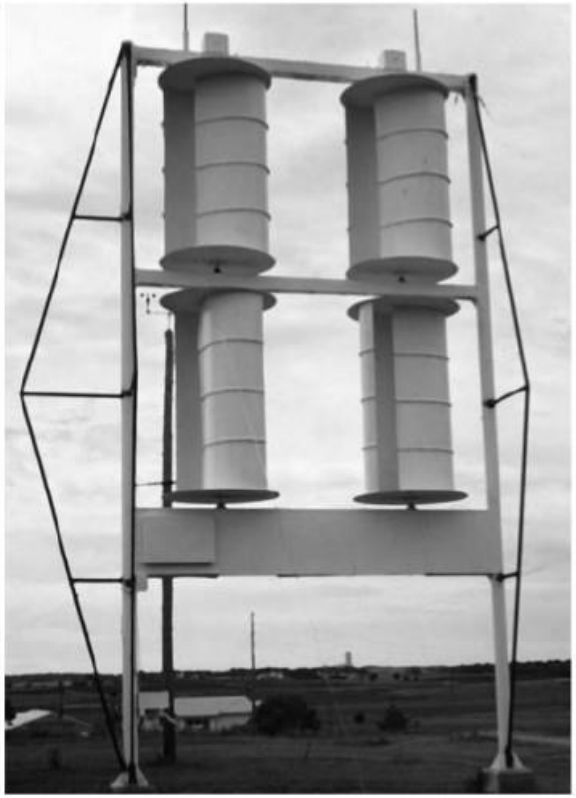
Categories of wind turbines

- Wind turbines can be divided into two categories namely; vertical axis wind turbine (VAWT) and horizontal axis wind turbine (HAWT).
- The two categories are based on the axis around which the turbine blades rotate.

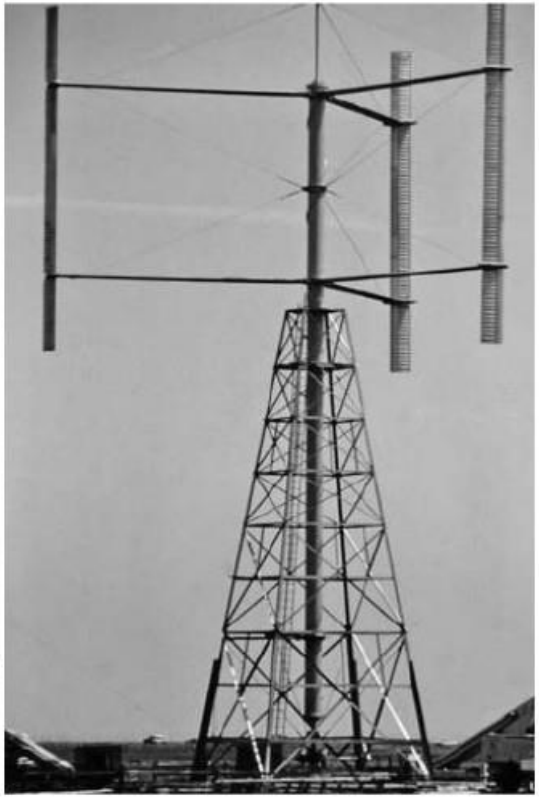
Vertical Axis Wind Turbine (VAWT)



Darrieus type

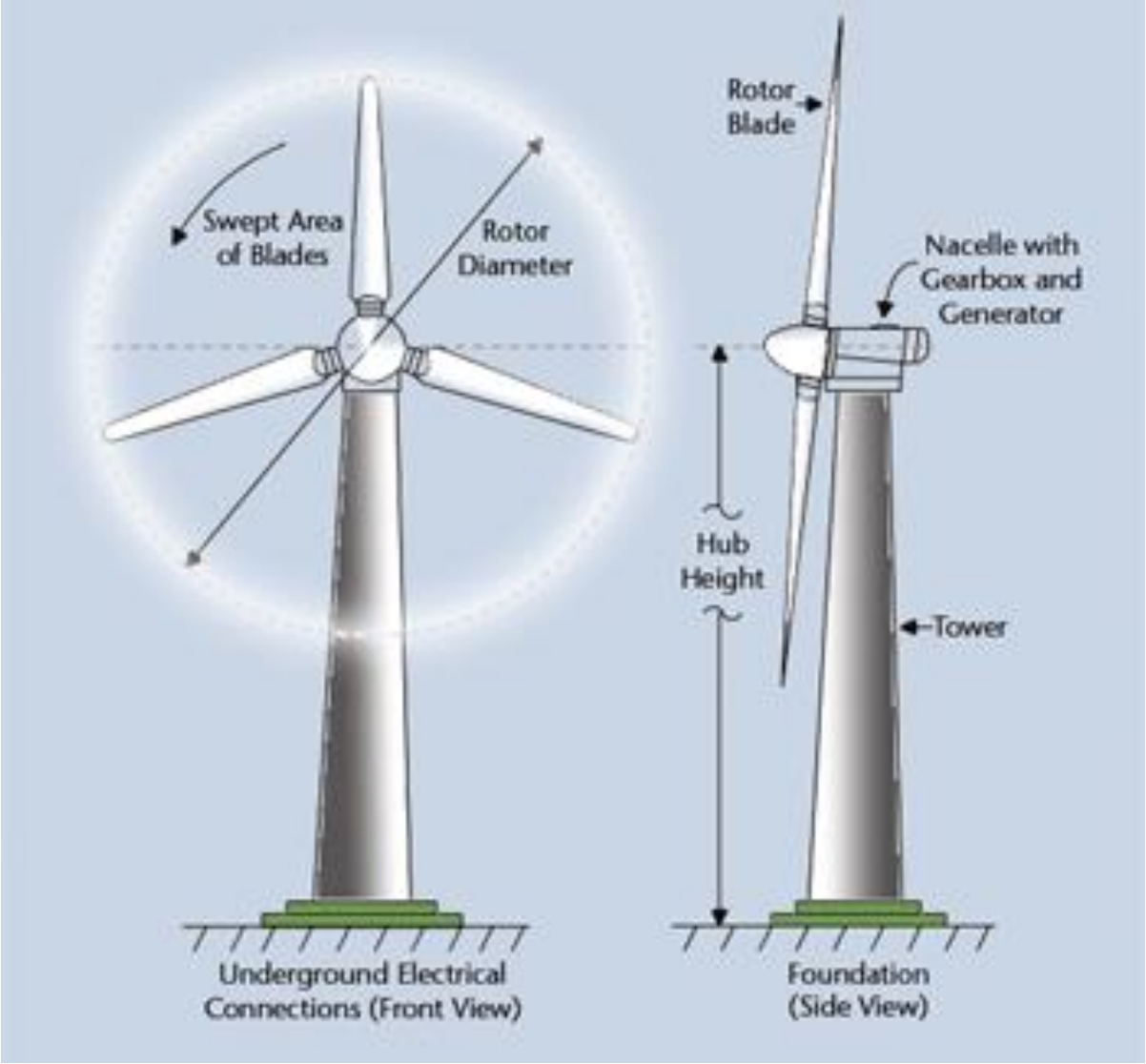


Savonius type

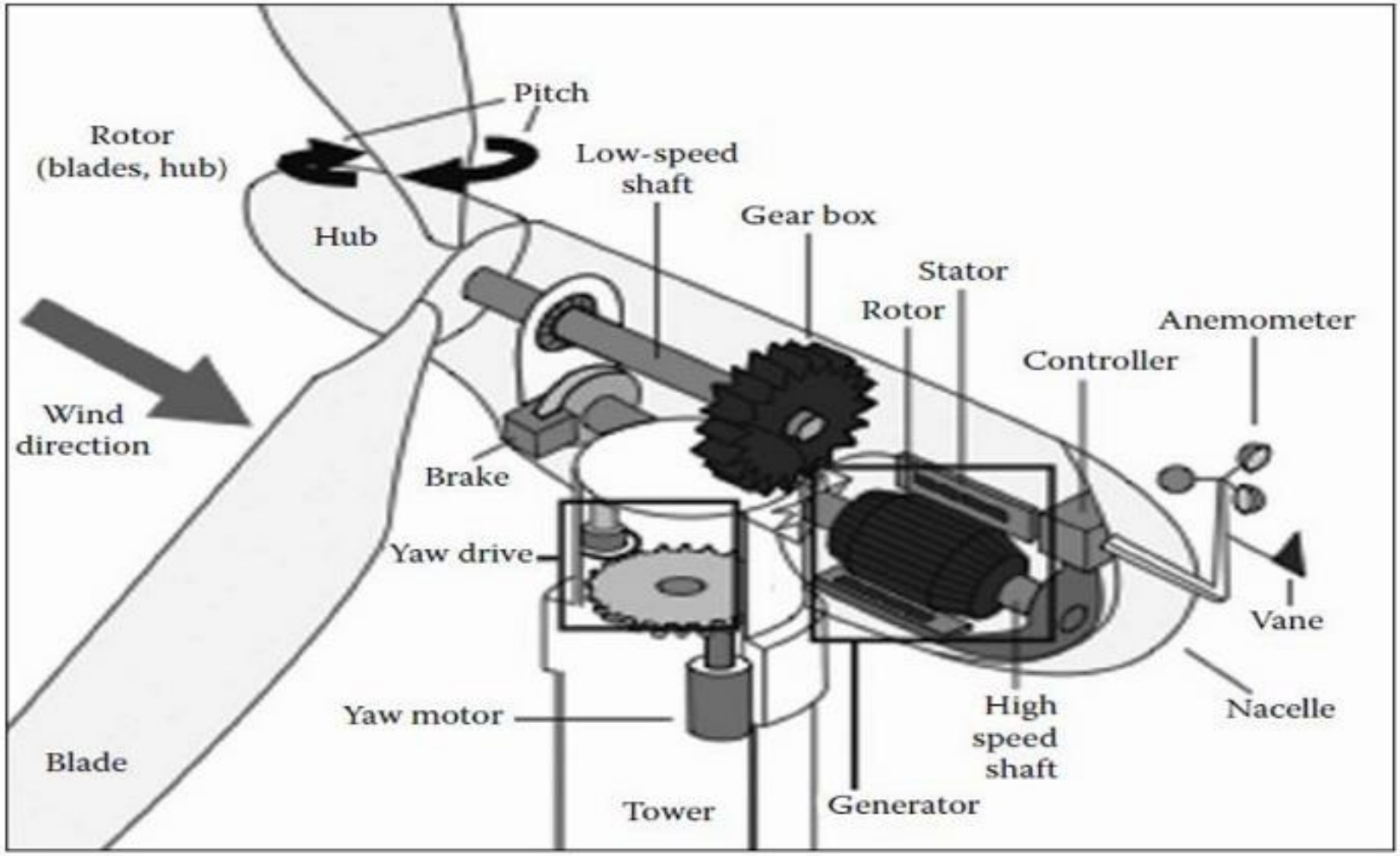


Giromill type

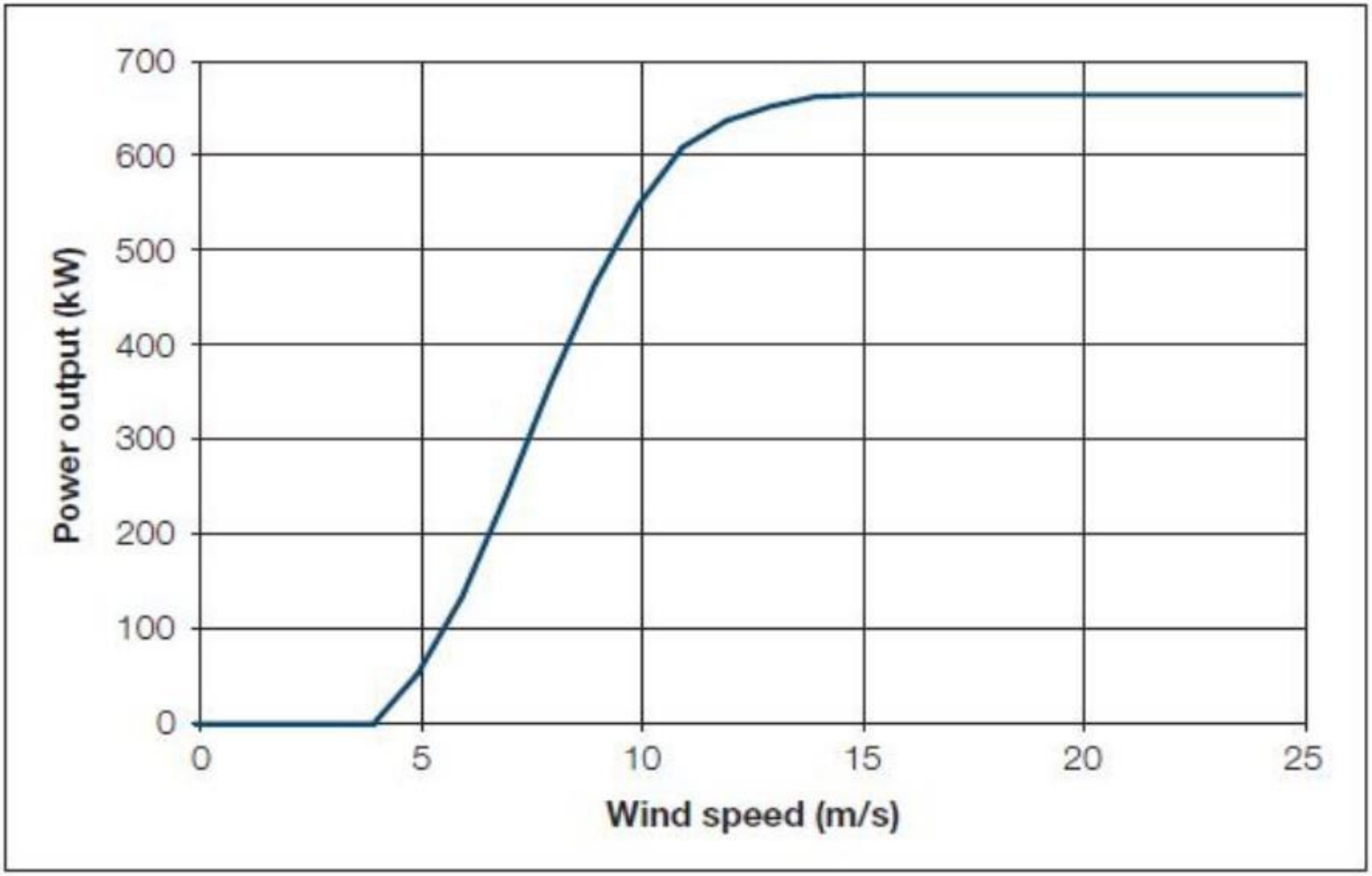
Horizontal Axis Wind Turbine (HAWT)



Components of a Horizontal Axis Wind Turbine



Wind power curve



Operational speeds of wind turbines

- *Start-up wind speed* – the wind speed that will turn an unloaded rotor
- *Cut-in wind speed* – the minimum wind speed at which power is generated
- *Rated wind speed* – the wind speed at which the wind turbine is designed to run to produce rated power
- *Cut-out wind speed* – the wind speed at which the wind turbine is programmed to stop in order to prevent damage
- *Maximum design wind speed* – the wind speed above which damage could occur to the wind turbine

Other wind power terminologies

- *Power Rating* - Wind turbines are typically defined by their power generating capacity. The rated power is produced at the rated wind speed of the turbine. The rated wind speed is usually a strong wind of about 15 m/s.
- *Capacity Factor* **-** It relates the actual energy output to the maximum possible output if the machine were running at its rated power during all of the 8760 hours of the year.

$$C_p = \frac{Actual\ annual\ energy\ output}{Maximum\ possible\ energy\ output}$$

Wind resource assessment

- First step in assessing wind power potential
- Used to predict energy production
- The wind speed varies by the minute, hour, day, season and year.
- Therefore, the annual mean wind speed needs to be averaged over ten or more years (quite expensive and, so a year's data could be used)

Wind Speed Measurement



Power Law

- The power law for wind shear is defined as:

$$\frac{U_h}{U_{ho}} = \left(\frac{h}{h_o}\right)^{\alpha}$$

- Where:
  - h is the height at which speed U_h is to be estimated.
  - ho is the height at which speed U_{ho} was measured.
  - α is the wind shear exponent or the friction coefficient.

Potential environmental and social impacts of wind power development

- Noise
- Bird Strikes
- Visual impacts
- Interference with telecommunication signals
- safety